

# Bayesian Decision Theory

## Exercise 1

A mobile robot is tasked with identifying and picking up *target objects* (e.g., wooden objects) while ignoring non-target objects. The object class is defined as

$$C \in \mathcal{C} = \{0, 1\},$$

where  $C = 1$  denotes a **target object**, and  $C = 0$  denotes a **non-target object**. The prior probabilities are given by  $p_C(0) = 0.3$ ,  $p_C(1) = 0.7$ .

The robot uses sensor readings (e.g., a camera) to classify objects. We built a machine learning model which takes the sensor readings, and return  $X = 1$  to tell us if the object is a target, and  $X = 0$  to tell us it is not.

For a target object ( $C = 1$ ), the probability that the machine learning model is correct is

$$p_{X|C}(X = 1 \mid C = 1) = 0.85,$$

and for a non-target object ( $C = 0$ ), the probability of a false positive (i.e., the machine learning model telling us that a non-target object is a target) is

$$p_{X|C}(X = 1 \mid C = 0) = 0.2.$$

Assume that the machine learning model is returning  $X = 1$  based on the sensor readings.

The available decision actions are:

- $a_0$ : Ignore the object (i.e., assume  $C = 0$ ).
- $a_1$ : Pick up the object (i.e., assume  $C = 1$ ).
- $a_{\text{reject}}$ : Re-measure the object and do another prediction (i.e., reject making a final decision).

The loss function is defined as follows:

$$L(c, a) = \begin{cases} 0, & \text{if } c = 0 \text{ and } a = a_0 \quad (\text{correct identification}), \\ 0, & \text{if } c = 1 \text{ and } a = a_1 \quad (\text{correct identification}), \\ 1, & \text{if } c = 0 \text{ and } a = a_1 \quad (\text{erroneous pick up}), \\ 1, & \text{if } c = 1 \text{ and } a = a_0 \quad (\text{missed target}), \\ \lambda, & \text{if } a = a_{\text{reject}}, \end{cases}$$

with  $\lambda = 0.1$ .

- (a) Compute the posterior probability  $p_{C|X}(c | x)$  given that  $X = 1$ .
- (b) Determine the expected risks  $R(a_0 | 1)$ ,  $R(a_1 | 1)$ , and  $R(a_{\text{reject}} | 1)$ .
- (c) Using the decision rule with a reject option,

$$a^*(1) = \begin{cases} a_{\arg\max_{c \in \mathcal{C}} p_{C|X}(c | 1)} & \text{if } \max_{c \in \mathcal{C}} p_{C|X}(c | 1) > 1 - \lambda, \\ a_{\text{reject}} & \text{otherwise,} \end{cases}$$

decide the optimal action  $a^*(1)$  for the robot.

- (d) Find  $a^*(1)$  but with  $\lambda = 0.4$  (higher cost on re-measuring).